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324316 (24)

BE (3rd Semester)

Examination, April - May, 2013

Branch : Electrical, EEE

ELECTRICAL POWER GENERATION

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Internal choice is given.

UNIT - I

Q. 1. Define demand factor. 2

Q. 2. A residential consumer has a connected load of 6 lamps each of 100 W and 4 fans each of 60 W at his premises. His demand is as follows : 14

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From mid night to 5 am	120 W
From 5 am to 6 pm	No load
From 6 pm to 7 pm	380 W
From 7 pm to 9 pm	680 W
From 9 pm to mid night	420 W

(i) Plot the load curve.

(ii) Calculate the demand factor, average load, maximum load and load factor.

OR

Q. 3. A generating station has a maximum demand of 100 MW, a load factor of 65%, a plant capacity

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factor of 50% and a plant use factor of 75%, find : 14

(i) The daily energy produced,

(ii) The reserve capacity of the plant,

(iii) The maximum energy that could be produced daily if the plant when running were fully loaded.

UNIT - II

Q. 1. Explain in short what is condenser ? 2

Q. 2. Explain the feed water and cooling water circuit in a steam power plant. 7

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Q. 3. Explain the function of the following in a steam power station : 7

(i) Economiser

(ii) Superheater

(iii) Condenser

(iv) Air preheater

OR

Q. 4. Draw a typical layout of steam power plant explain in brief main feactures of layout. 7

UNIT - III

Q. 1. What is hydrology ? 2

Q. 2. Explain the principle of working of a hydro electric power plant. 7

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Q. 3. Discuss different types of dams used in hydro power plant. 7

OR

Q. 4. Explain the following terms : 7

(i) Run-off

(ii) Pumped storage plant

UNIT - IV

Q. 1. What is nuclear chain reaction ? 2

Q. 2. Draw and explain the function of main parts of a nuclear reactor. 7

Q. 3. Classify nuclear reactor and explain with a neat diagram, the pressurized water reactor. 7

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OR

Q. 4. Explain the principle of MHD generation. Also explain the closed cycle MHD system with the help of a neat sketch. 7

UNIT - V

Q. 1. Name different types of non-conventional energy sources. 2

Q. 2. What is the criteria for distribution of load between units of a plant ? Explain transmission loss as a function of plant generation. 7

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Q. 3. Give the conversion of solar energy into electrical energy. How photo voltaic cell work. 7

OR

Q. 4. Explain solar hydrogen energy cycle. 7

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